MIL-STD-189
15 November 1955

MILITARY STANDARD

RACKS, ELECTRICAL EQUIPMENT, 19-INCH AND ASSOCIATED PANELS

UNITED STATES

WASHINGTON, D. C.-1955

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OFFICE OF THE ASSISTANT SECRETARY OF DEFENSE
WASHINGTON 25, D. C.

15 November 1955

Supply and Logistics

RACKS, ELECTRICAL EQUIPMENT, 19-INCH AND ASSOCIATED PANELS

MIL–STD–189

1. This standard has been approved by the Department of Defense for use by the Departments of the Army, the Navy, and the Air Force.

2. In accordance with established procedure, the Standardization Division has designated the Signal Corps, the Bureau of Ships, and the Air Force, respectively, as Army-Navy-Air Force custodians of this standard.

3. This standard is mandatory for use effective 15 November 1955 by the Departments of the Army, the Navy, and the Air Force.

4. Recommended corrections, additions, or deletions should be addressed to the Standardization Division, Office of the Assistant Secretary of Defense (Supply and Logistics), Washington 25, D. C.

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RACKS, ELECTRICAL EQUIPMENT, 19 INCH AND ASSOCIATED PANELS

1. SCOPE

1.1 This standard established the form factor for 19-inch racks and panels, and those dimensions which are critical in ensuring interchangeable use of these items. It is intended only as a guide for preparing procurement specifications for the racks and panels, and equipment to be mounted thereon.

2. DEFINITIONS

2.1 **Rack**—The rack covered by this standard is a non-enclosed framework having uprights of channel-shaped section and attached base.

3. GENERAL REQUIREMENTS

3.1 **Construction.**—The frame shall be constructed as a rigid unit, made up of two channels, two straps and one angle. These parts shall be permanently secured to form a rigid structure, using the best available methods for the materials involved. The base and brackets shall be supplied disassembled from the frame.

3.2 **Dimensions.**

3.2.1 *Rack heights.* — Racks shall be in accordance with heights given in table I, except that overall rack heights, including shockmounts and base if any, shall not exceed 72 inches for shipboard use and 60 inches for van and shelter use as specified by the bureau or agency concerned. Table I specifies nominal values.

TABLE I. *Nominal rack heights*

Rack size	Frame height	Panel clearance height ±1/64
	Inches	*Inches*
A	$21\frac{21}{32}$	$17\frac{21}{32}$
B	$28\frac{21}{32}$	$24\frac{21}{32}$
C	$35\frac{21}{32}$	$31\frac{21}{32}$
D	$42\frac{21}{32}$	$38\frac{21}{32}$
E	$49\frac{21}{32}$	$45\frac{21}{32}$
F	$56\frac{21}{32}$	$52\frac{21}{32}$
G	$63\frac{21}{32}$	$59\frac{21}{32}$
H	$70\frac{21}{32}$	$66\frac{21}{32}$
J	$77\frac{21}{32}$	$73\frac{21}{32}$
K	$84\frac{21}{32}$	$80\frac{21}{32}$

3.2.2 *Panel dimensions.*—Panel thickness shall be 1/8, 3/16, 1/4, or 5/16 inch, as specified in the individual contract or order. Other panel dimensions in inches with their letter designations (AO, BO, and BC) shall be as specified in table II. Dimensions have been chosen to allow a nominal clearance of 1/32 inch between panels on any one rack.

TABLE II. *Panel dimensions other than thicknesses*

Panel size		A	B	C	D	E	F	G	H
Open slot	Closed slot								
AO	AC	$1\frac{23}{32}$	$1\frac{1}{4}$	-----	-----	-----	-----	-----	-----
BO	BC	$3\frac{15}{32}$	3	-----	-----	-----	-----	-----	-----
CO	CC	$5\frac{7}{32}$	$2\frac{1}{4}$	-----	-----	-----	-----	-----	-----
DO	DC	$6\frac{31}{32}$	4	-----	-----	-----	-----	-----	-----
EO	EC	$8\frac{23}{32}$	$1\frac{3}{4}$	$2\frac{1}{4}$	$1\frac{3}{4}$	-----	-----	-----	-----
FO	FC	$10\frac{15}{32}$	$2\frac{1}{4}$	3	$2\frac{1}{4}$	-----	-----	-----	-----

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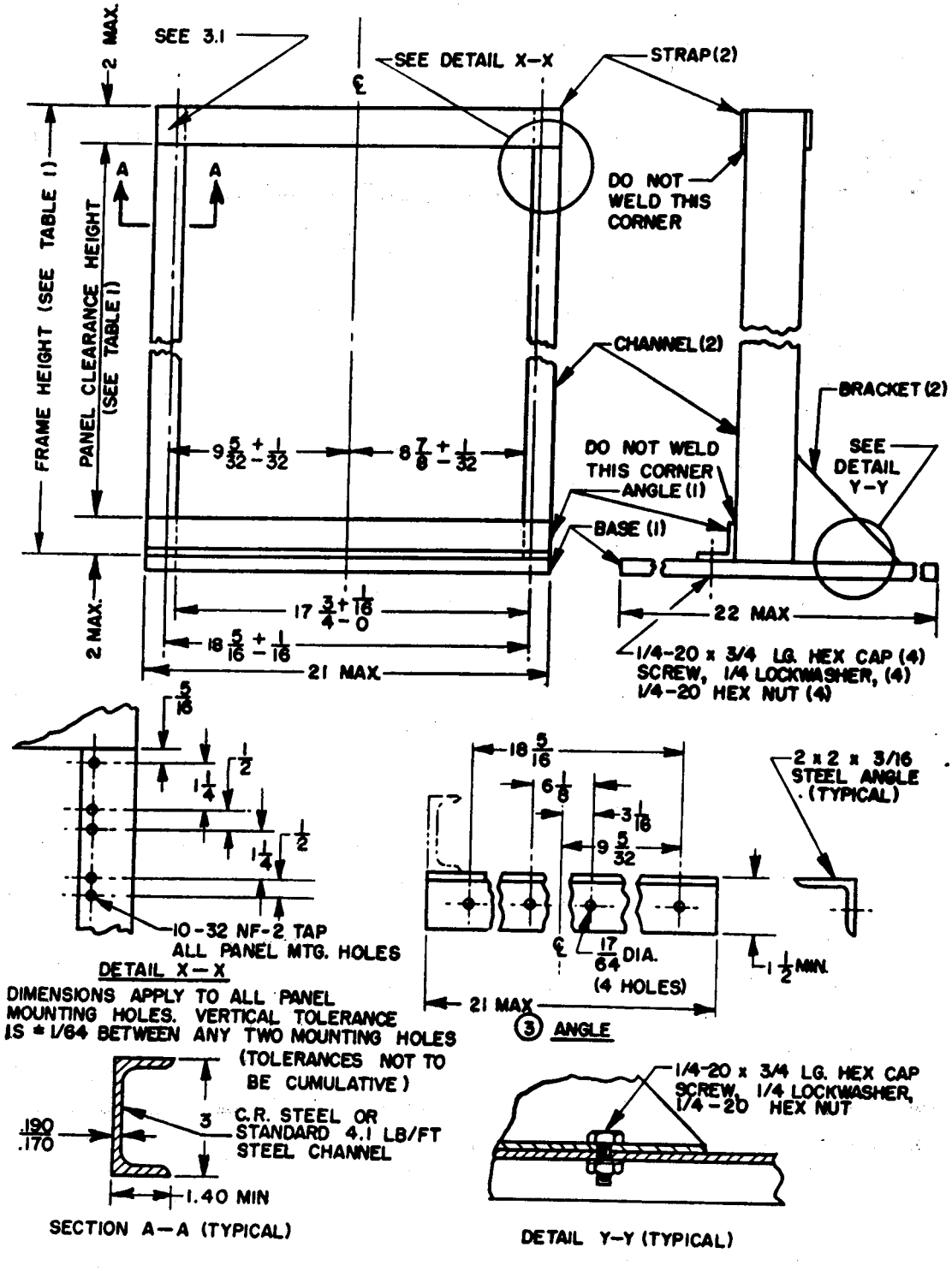
TABLE II. *Panel dimensions other than thicknesses, continued*

Panel size		A	B	C	D	E	F	G	H
Open slot	Closed slot								
GO	GC	$12\frac{7}{32}$	1¾	5¾	1¾	.....	.....	.....	.....
HO	HC	$13\frac{31}{32}$	3½	4	3½	.....	.....	.....	.....
JO	JC	$15\frac{23}{32}$	3½	5¾	3½	.....	.....	.....	.....
KO	KC	$17\frac{15}{32}$	3½	7½	3½	.....	.....	.....	.....
LO	LC	$19\frac{7}{32}$	5¼	5¾	5¼	.....	.....	.....	.....
MO	MC	$20\frac{31}{32}$	1¾	5¼	4	5¼	1¾	.....	.....
NO	NC	$22\frac{23}{32}$	5¼	1¾	5¾	1¾	5¼	.....	.....
PO	PC	$24\frac{15}{32}$	1¾	5¼	7½	5¼	1¾	.....	.....
RO	RC	$26\frac{7}{32}$	1¾	7	5¾	7	1¾	.....	.....
SO	SC	$27\frac{31}{32}$	1¾	5¼	3½	4	3½	5¼	1¾
TO	TC	$29\frac{23}{32}$	1¾	5¼	3½	5¾	3½	5¼	1¾
UO	UC	$31\frac{15}{32}$	1¾	5¼	3½	7½	3½	5¼	1¾

Note.—Vertical tolerance is $\pm\frac{1}{64}$ between any two slots (tolerances not to be accumulative).

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3.2.3 *Limiting dimensions of racks.*—Limiting dimensions of racks shall conform to figure 1.



NOTE: Drawings marked "typical" illustrate form factors only; they represent one of several possible assembly or fabrication techniques.

Figure 1.—Limiting dimensions for 19-inch electrical equipment racks.

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3.2.4 *Limiting dimensions of panels.*—Limiting dimensions of panels shall conform to figure 2.

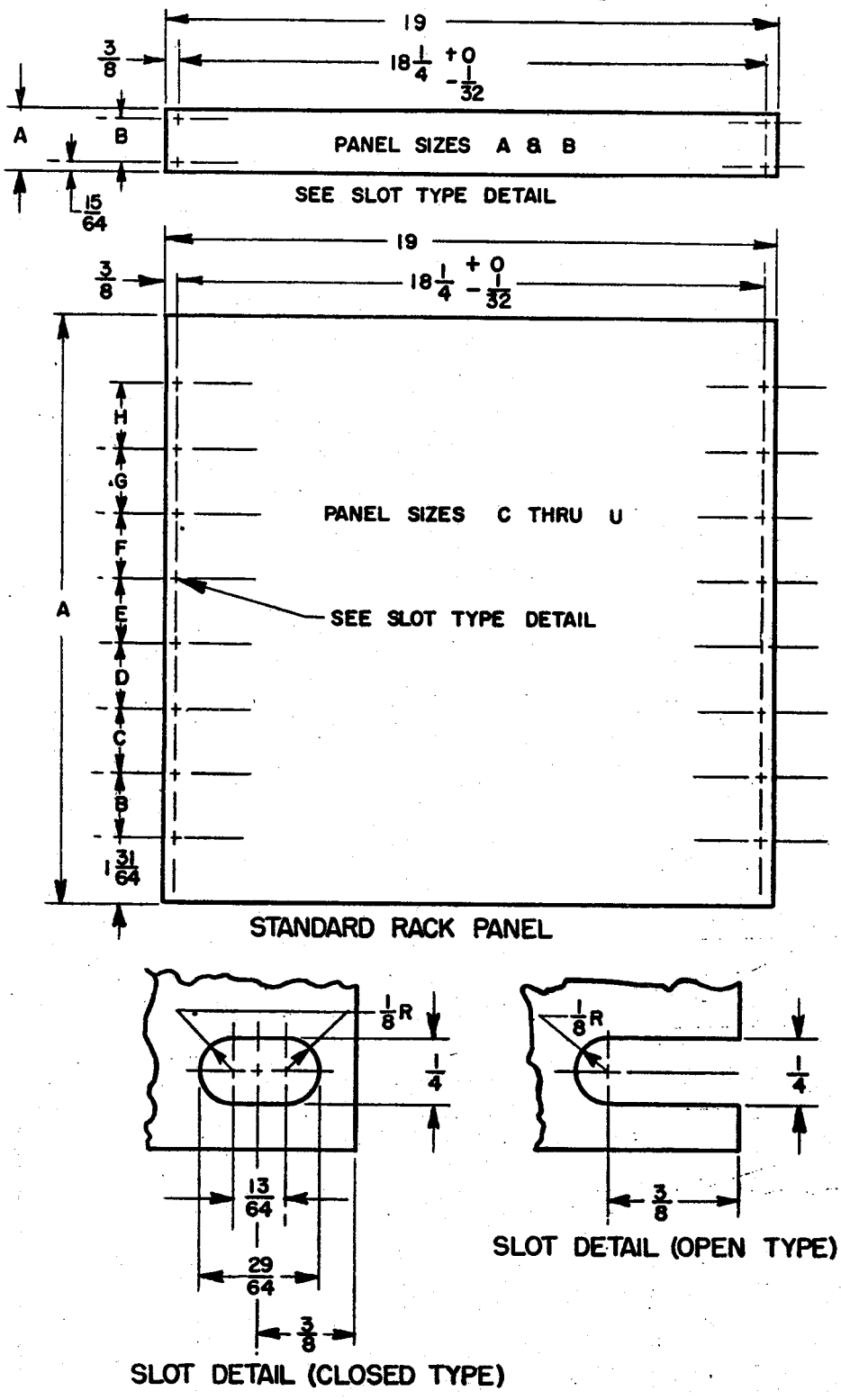


Figure 2.—Limiting dimensions for 19-inch electrical equipment rack panels.

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